

ASUTOSH PANDA

📍 Brahmapur, India • ✉ asutosh.pda@gmail.com • 🌐 75asu.pages.dev • 🐙 github.com/75asu • 🔗 linkedin.com/in/75asu

SKILLS

GPU & AI Infrastructure: NVIDIA GPU Operator, DCGM, drivers/CUDA, MIG, vLLM inference serving, GPU SLOs and observability
Infrastructure & Networking: Kubernetes (GKE, bare-metal), Helm, Kustomize, FluxCD, ArgoCD, Terraform, OpenTofu, Ansible, Linkerd, KubeOne, OpenEBS, GCP, AWS, Hetzner, OVH, Cloudflare
Secrets and Auth: GCP Workload Identity Federation, External Secrets Operator (ESO), GCP Secret Manager, cert-manager, Pomerium
CI/CD and Observability: GitHub Actions, self-hosted ARC runners, Trivy, SonarQube, TruffleHog, Renovate; Prometheus, Grafana, Mimir, Loki, Tempo, Grafana Alloy, grafana-operator, Alertmanager
Development: Golang, Python, FastAPI, Bash, Node.js, TypeScript, PostgreSQL, Linux

PROFESSIONAL EXPERIENCE

ONE2N

Pune, Maharashtra, India

Senior SRE & Team Lead • Client: Fravity AI (Remote)

May 2026 - Present

- Built **fravity-spine**, the production multi-tenant GKE platform powering an AI product: private control plane, CMEK etcd, Workload Identity Federation, OpenTofu across **6** isolated state layers. Replaced bastions and shared SSH with identity-based **kubectrl** via GKE Connect Gateway + Workspace SSO; FluxCD-bootstrapped cert-manager, ESO, Traefik, and Pomerium SSO.
- Led a full-estate infrastructure assessment spanning VMs and **5** GCP plus **5** AWS accounts, cataloguing security, IAM, and cost risks into a prioritised remediation roadmap: surfaced **19** service-account keys, an unowned **32-vCPU** instance, and ~**\$26,520/yr** of zombie spend. Drove the follow-up hardening: IaC bootstrapped to **60%** across **2** envs, **10** workload SAs and all human IAM migrated to Terraform, CI rebuilt on Workload Identity Federation. Built and maintained the platform roadmap and backlog from these findings, prioritising and distributing the highest-impact work across the **2-engineer** team I lead.
- Delivered the central LGTM observability platform: Mimir + Loki + Tempo (GCS-backed via Workload Identity, **365-day** metrics/logs + **30-day** traces, multi-tenant via X-Scope-OrgID) as authenticated public ingest. Migrated to grafana/alloy single-agent (logs, metrics, traces) across spine and the fravity-dev1 spoke; Alertmanager + Slack via ESO; grafana-operator CRDs for dashboards-as-code.
- Stood up a **5-tool** SAST stack on self-hosted ARC runners (Trivy operator, SonarQube CE, renovate-operator, TruffleHog) to meet the "scans inside our private GKE" audit bar. Shipped **scan-aggregator**, a FastAPI webhook service normalising **7** scanner outputs to a per-repo audit **xlsx**; first customer-facing deliverable: a customer security review with **76** H/C findings across **4** images.
- Established the platform's first formal on-call and incident process (primary/secondary rotation, SEV1 runbooks, weekly executive status to the CEO), and raised team maturity by mentoring developers and standardising team tooling (Outline docs, Vaultwarden secrets).
- Delivered staging on **fravity-dev1** GKE as a commit-to-URL GitOps pipeline: CI build, Flux image-automation HelmRelease bumps, Atlas migration gate, a live TLS endpoint, and Slack deploy alerts, all **zero static credentials** via Workload Identity with per-app least-privilege GSAs. Root-caused the rollout's stacked failures (HPA runaway, cgroup OOMs, DB networking) and upstreamed the chart fix.

LUXOR TECHNOLOGY

Seattle, WA, US

DevOps Engineer (Remote)

June 2023 - April 2026

- Operated a multi-cluster Kubernetes platform for **20+** teams and **3,000+** enterprise customers: **8+** clusters across Hetzner, OVH, and GCP with **24/7** uptime for L1/L2 validator and RPC workloads (BTC, LTC, Zcash, Stacks, tBTC), including **2** Hetzner bare-metal HA clusters (KubeOne, OpenEBS) provisioned and driver-managed from scratch. Standardised on a **7-layer** FluxCD GitOps pipeline with Linkerd mTLS; migrated **70+** SOPS secrets to GCP Workload Identity.
- Led incident response for a Litecoin mainnet hard fork: cross-team build, **zero-downtime** rolling deployment across production, postmortem. Restored production after a kubelet TLS expiry on bare metal; deployed x509 monitoring across **10** clusters with **7-day/24-hour** PromQL alerts.
- Owned CI/CD for **50+** repositories on self-hosted ARC runners. Ported a failing Playwright E2E suite from Mac hosted runners (**\$5,000/mo**) to Linux ARC, cutting runtime **40 to 15 min** and cost to **\$600/mo**. Built code-signed Windows + Mac release pipelines and automated FluxCD deployments to GCP Artifact Registry.
- Built org-wide platform tooling: **KubeOrbit** (Go CLI for kubeconfig provisioning across **20+** clusters) and **Memphis** (Python SOC1/2/SOX compliance binary with auditor-ready Drive reports + Linear escalation; solo, **4 weeks** for the planned IPO). Prevented a supply-chain compromise by vendoring Trivy into a composite action across **9** repos; when CVE-2026-33634 later hit aquasecurity/trivy-action, Luxor pipelines were untouched.
- Proposed and built **BVT**, the org's first zero-hardcoded-secrets cluster: replaced SOPS-in-git with ESO + GCP Secret Manager + Workload Identity Federation; the pattern became the gold standard for new clusters. Authored custom Python tooling against Hetzner + OVH firewall APIs to enable secure cross-cloud peering for bare-metal nodes where no Terraform modules existed.

TATA CONSULTANCY SERVICES

Hyderabad, Telangana, India

Systems Engineer (Remote)

October 2020 - June 2023

- Migrated **2,500+** repositories from GitLab, Bitbucket, and SVN to GitHub Enterprise for **120** teams, saving **15 weeks** of manual effort.
- Built PowerShell, Python, and Golang automation across **22** RHEL/WebLogic servers (health checks and storage allocation), saving **380+** man-hours/month.

PROJECTS

- Frit** (May 2026 - Present) 📄 - GPU SRE lab: vLLM on NVIDIA GPU Operator; OSS PRs to **dcgm-exporter** + **gpu-operator**.
- agent-box** (One2N, 2026) - Hardened devcontainer sandbox for AI agents: cap-drop ALL, non-root, no host credentials.
- Platform Zero** (Apr 2026 - Present) 📄 - **4** SRE labs (k8slab, ansiblelab, awslab, gcplab); **0-to-100** in one command.
- Truss** + **Dint** (2026) - Self-hostable BaaS (auth, storage, realtime) and an embeddable user-feedback widget.

TEACHING EXPERIENCE

- Full-Stack Instructor, **4** cohorts (2022-2026), taught NextJS, FastAPI, and DevOps to BSc CS students.

EDUCATION

SASTRA DEEMED UNIVERSITY

Master of Computer Applications (MCA) - Chennai, India - 2021 - 2023